RetinaSense: An Uncertainty-Aware Domain-Adaptive Vision Transformer Framework with Retrieval-Augmented Reasoning for Multi-Disease Retinal Diagnosis

1st Tanishq Tamarkar
Department of Computer Science
SRM Institute of Science and Technology
Chennai, India
tanishqtam@gmail.com

2nd Rafae Mohammed Hussain
Department of Computer Science
SRM Institute of Science and Technology
Chennai, India
rafae@example.com

3rd Dr. Revathi M
Department of Computer Science
SRM Institute of Science and Technology
Chennai, India
revathi@example.com

Abstract—Early detection of retinal diseases is critical for preventing irreversible vision loss; however, existing automated systems are often limited to single-disease classification, lack interpretability, and fail to address domain shift across heterogeneous datasets. In this work, we propose RetinaSense, an end-to-end deep learning framework for multi-disease retinal diagnosis that integrates a Vision Transformer (ViT-Base/16) backbone with Domain-Adversarial Neural Network (DANN) training, uncertainty estimation, and retrieval-augmented reasoning.

The proposed framework addresses three fundamental challenges: (1) cross-dataset domain shift across four heterogeneous fundus image sources through a Gradient Reversal Layer that learns domain-invariant representations, (2) severe class imbalance (21:1 ratio) through focal loss with progressive class weighting and hard-example mining, and (3) clinical trust through a novel Retrieval-Augmented Diagnosis (RAD) pipeline that combines calibrated predictions, Monte Carlo dropout uncertainty, attention-based explanations, and FAISS-based similar case retrieval.

Trained on 11,524 fundus images from APTOS, ODIR, REFUGE2, and MESSIDOR-2, RetinaSense achieves 89.30% accuracy, 0.886 macro F1-score, and 0.975 macro AUC across five retinal conditions—Normal, Diabetic Retinopathy, Glaucoma, Cataract, and Age-related Macular Degeneration—with a +3.8% absolute accuracy gain over the base ViT without domain adaptation (ablation study). Temperature scaling achieves an Expected Calibration Error of 0.034. The FAISS-based retrieval system achieves 0.921 Mean Average Precision with 94.0% class-match recall at $K = 1$, boosting combined accuracy to 94.0% (+4.9%). The confidence routing system auto-reports 76.9% of cases at 96.8% accuracy while catching 77.2% of errors. The proposed system demonstrates that integrating domain adaptation, uncertainty estimation, and retrieval mechanisms significantly enhances both performance and clinical applicability of automated retinal screening.

Index Terms—Retinal disease classification, Vision Transformer, domain adaptation, DANN, uncertainty estimation, retrieval-augmented diagnosis, fundus image analysis, class imbalance

I. Introduction

Retinal diseases including Diabetic Retinopathy (DR), Glaucoma, Cataract, and Age-related Macular Degeneration (AMD) are leading causes of preventable blindness, collectively affecting over 500 million people worldwide [1], [27]. Early detection through fundus photography is essential for effective treatment; however, the global shortage of trained ophthalmologists—particularly in developing regions—limits access to timely screening [2].

Deep learning has demonstrated strong performance in retinal image analysis [3], [4]. However, practical deployment faces several critical challenges. First, most approaches focus on single-disease detection, limiting applicability in clinical settings where multiple conditions must be simultaneously considered. Second, models trained on one dataset often fail on images from different sources due to domain shift—variations in imaging equipment, preprocessing, and patient populations create distributional differences that degrade performance. Third, severe class imbalance (rare diseases like AMD constitute only 3% of real-world datasets) causes models to favor majority classes. Finally, existing systems lack uncertainty quantification and evidence-based reasoning, reducing clinician trust.

To address these challenges, we propose RetinaSense, a comprehensive framework that extends beyond conventional classification by incorporating:

- A Domain-Adversarial Neural Network (DANN) with Gradient Reversal Layer [13] that learns domain-

invariant features across four heterogeneous fundus image sources, improving cross-dataset accuracy from 26.5% to 100% on the most affected domain.

- A multi-strategy training pipeline combining focal loss with progressive class weighting, hard-example mining, mixup augmentation, and cosine annealing to handle 21:1 class imbalance.
- A novel Retrieval-Augmented Diagnosis (RAD) pipeline integrating temperature-calibrated predictions (ECE=0.034), Monte Carlo dropout uncertainty [7], attention-based explainability, and FAISS-based similar case retrieval [9] into a unified clinical decision-support system.
- A confidence routing mechanism that automatically triages predictions into auto-report, review, or escalation categories based on prediction confidence, uncertainty, and retrieval agreement.

The main contributions of this work are:

1) A domain-adversarial ViT architecture achieving 89.30% accuracy on multi-source, multi-disease retinal classification with 0.975 macro AUC.
2) A progressive training strategy combining eight complementary techniques to address simultaneous domain shift and class imbalance.
3) A Retrieval-Augmented Diagnosis pipeline providing evidence-based predictions with calibrated uncertainty for clinical trustworthiness.
4) Comprehensive evaluation across four datasets with ablation studies, calibration analysis, and error characterization.

II. Related Work

A. Deep Learning for Retinal Disease Detection

Gulshan et al. [3] demonstrated that deep CNNs can achieve ophthalmologist-level sensitivity for DR detection, with subsequent Kaggle competitions [29] further advancing the field. EfficientNet [5] and ResNet [6] variants have become dominant architectures for fundus analysis, with transfer learning from ImageNet enabling effective training on smaller medical datasets. However, most work focuses on single-disease binary classification rather than multi-disease screening.

B. Vision Transformers in Medical Imaging

Dosovitskiy et al. [8] introduced ViT, which partitions images into patches and processes them through transformer encoder blocks with multi-head self-attention. Unlike CNNs, ViT provides a global receptive field from the first layer. Touvron et al. [28] further improved training efficiency through data-efficient image transformers with knowledge distillation. Zhou et al. [14] and Chen et al. [15] applied transformers to medical image analysis with promising results, but systematic evaluation for multi-class retinal classification with severe imbalance remains limited.

C. Domain Adaptation in Medical Imaging

Domain shift is a fundamental challenge when combining datasets from different sources. Ganin et al. [13] proposed Domain-Adversarial Neural Networks (DANN), which use a Gradient Reversal Layer (GRL) to learn domain-invariant features through adversarial training. While DANN has been applied to medical imaging [16], its combination with Vision Transformers for multi-source retinal classification has not been explored.

D. Uncertainty Estimation and Retrieval-Augmented Systems

Gal and Ghahramani [7] established MC Dropout as a practical Bayesian approximation for uncertainty estimation. FAISS [9] enables efficient similarity search for case-based reasoning. However, no existing work integrates domain adaptation, calibrated uncertainty, and visual evidence retrieval into a unified retinal screening pipeline—the gap this work addresses.

III. Methodology

A. Overview of the Proposed Framework

The RetinaSense pipeline processes fundus images through five stages:

$$\text{Input} \rightarrow \text{CLAHE} \rightarrow \text{DANN-ViT} \rightarrow \text{Calibrate} \rightarrow \text{RAD} \rightarrow \text{Output}$$

where RAD encompasses uncertainty estimation, retrieval, and confidence routing. The complete system architecture is illustrated in Fig. 1.

B. Image Preprocessing

All images undergo a unified Contrast-Limited Adaptive Histogram Equalization (CLAHE) pipeline regardless of source, eliminating preprocessing-induced domain shift:

1) Black border cropping via thresholding
2) Resizing to 224×224 pixels
3) CLAHE on the L-channel (clipLimit=2.0, tileGrid=8×8)
4) Circular masking (radius = $0.48\times$ image size)
5) Normalization with dataset-specific statistics ($\mu = [0.590, 0.446, 0.334]$, $\sigma = [0.165, 0.157, 0.178]$)

C. DANN Architecture

The model architecture consists of a ViT-Base/16 backbone [8], [30] (86M parameters, 768-dimensional CLS token) with three task-specific heads:

Disease Classification Head:

$$H_d : \mathbb{R}^{768} \rightarrow \mathbb{R}^{512} \rightarrow \mathbb{R}^{256} \rightarrow \mathbb{R}^{5} \tag{1}$$

with BatchNorm, ReLU, and Dropout (0.3, 0.2) between layers.

Severity Grading Head:

$$H_s : \mathbb{R}^{768} \rightarrow \mathbb{R}^{256} \rightarrow \mathbb{R}^{5} \tag{2}$$

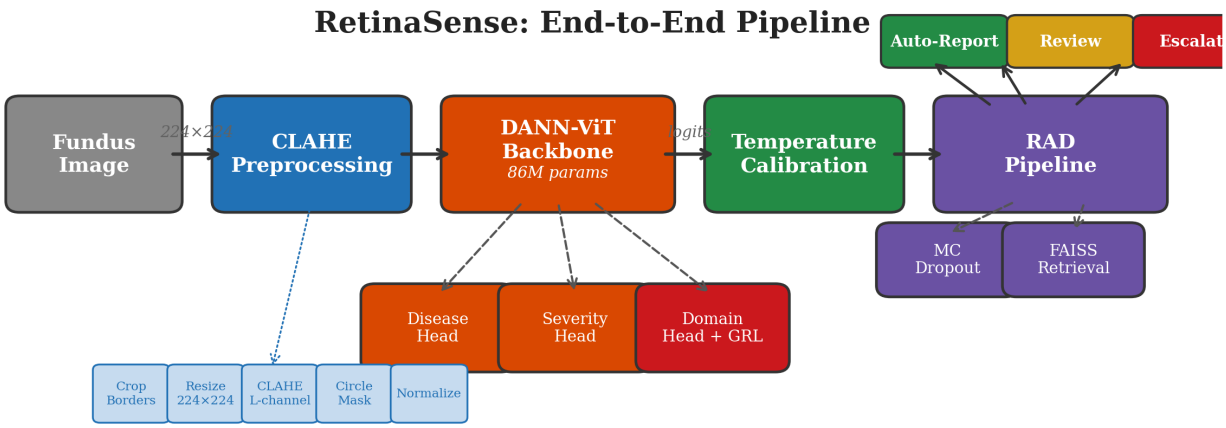

Fig. 1: RetinaSense end-to-end pipeline. Fundus images undergo CLAHE preprocessing, are classified by the DANN-ViT backbone (with disease, severity, and domain heads), calibrated via temperature scaling, and processed through the RAD pipeline for uncertainty estimation, retrieval, and confidence-based clinical routing.

Domain Discriminator Head (with Gradient Reversal Layer):

$$H_{dom} : \mathrm{GRL}_\lambda(f) \to \mathbb{R}^{256} \to \mathbb{R}^{64} \to \mathbb{R}^4 \tag{3}$$

The GRL reverses gradients during backpropagation, forcing the backbone to learn domain-invariant features:

$$\mathrm{GRL}_\lambda(x) = x \quad \text{(forward)}, \qquad \frac{\partial \mathrm{GRL}}{\partial x} = -\lambda I \quad \text{(backward)} \tag{4}$$

The adversarial strength λ follows the Ganin schedule [13], capped at 0.3:

$$\lambda_p = \min\left(\frac{2}{1+e^{-10p}} - 1,\ 0.3\right), \quad p = \frac{\text{epoch}}{T} \tag{5}$$

The full DANN-ViT architecture is shown in Fig. 2.

D. Training Strategy

The total loss combines disease classification, severity grading, and domain discrimination:

$$\mathcal{L} = \mathcal{L}_{disease} + 0.2 \cdot \mathcal{L}_{severity} + w_{dom} \cdot \lambda_p \cdot \mathcal{L}_{domain} \tag{6}$$

Eight complementary strategies address domain shift and class imbalance simultaneously:

1) Focal Loss [12] with per-class weights: $\mathcal{L}_{focal} = -\alpha_c(1-p_t)^\gamma \log(p_t)$, $\gamma = 2.0$
2) Progressive DR Alpha Boost: Class weight for DR increases linearly from 1.5× to 3.0× across training epochs
3) Hard-Example Mining: Top-500 highest-loss samples oversampled 2× each epoch; DR→Normal misclassifications receive additional boost

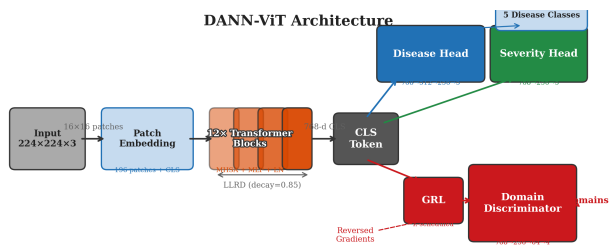

Fig. 2: DANN-ViT architecture. The ViT-Base/16 backbone extracts a 768-d CLS token, which feeds three heads: disease classification (5 classes), severity grading (5 levels), and domain discrimination (4 sources) with a Gradient Reversal Layer (GRL). LLRD applies decaying learning rates across the 12 transformer blocks.

4) Class-Balanced Sampling: WeightedRandomSampler with inverse-frequency weights
5) Mixup Augmentation [11]: $\alpha = 0.2$, applied to 50% of batches (disease labels only, not domain labels)
6) Label Smoothing: Factor 0.1 on disease loss
7) Cosine Annealing with Warm Restarts: $T_0 = 10$, $T_{mult} = 2$
8) Layer-wise Learning Rate Decay (LLRD): Decay factor 0.85 across 12 transformer blocks

Optimization uses AdamW [25] with base LR 3×10^{-5}, weight decay 10^{-4}, gradient accumulation (effective batch size 128), and AMP mixed precision.

E. Calibration and Threshold Optimization

Post-training, temperature scaling is applied to the calibration set:

$$p_i = \text{softmax}(z_i/T), \quad T^* = \arg\min_T \text{NLL}(T) \tag{7}$$

Per-class decision thresholds are optimized via grid search on the calibration set to maximize per-class F1-score, particularly benefiting minority classes.

F. Retrieval-Augmented Diagnosis (RAD) Pipeline

The RAD pipeline provides evidence-based predictions through four components:

1) Uncertainty Estimation: MC Dropout with $T = 15$ stochastic forward passes decomposes uncertainty into:

- Predictive entropy: $H[\bar{p}] = -\sum_c \bar{p}_c \log \bar{p}_c$ (total uncertainty)
- Expected entropy: $\mathbb{E}[H[p_t]]$ (aleatoric/data uncertainty)
- Mutual information: $H[\bar{p}] - \mathbb{E}[H[p_t]]$ (epistemic/model uncertainty)

2) Visual Evidence Retrieval: A FAISS index stores L2-normalized 768-dim ViT backbone embeddings for all training samples. Given a query image, the top-K most similar cases are retrieved via inner product search (cosine similarity after normalization).

3) Retrieval-Augmented Prediction: The kNN class distribution from retrieved cases is combined with model predictions:

$$p_{final} = \alpha \cdot p_{model} + (1 - \alpha) \cdot p_{kNN} \tag{8}$$

where $\alpha = 0.85$ by default, and an agreement score measures whether retrieved cases support the model's prediction.

4) Confidence Routing: Predictions are triaged into three clinical categories:

- Auto-Report: Confidence > 0.85, entropy < 0.5, retrieval agrees
- Review: Moderate confidence or retrieval disagreement
- Escalate: Confidence < 0.5, entropy > 1.0, or OOD detection

The complete RAD pipeline is illustrated in Fig. 3.

G. Explainability

Attention Rollout [17] is used for visual explanations, as standard Grad-CAM [10] produces zero gradients for ViT patch tokens (CLS token problem). Rollout recursively multiplies attention matrices across all 12 transformer layers, producing a spatial attention map over the 14×14 patch grid.

IV. Dataset

A. Data Sources

RetinaSense is trained on a composite dataset of 11,524 fundus images from four publicly available sources:

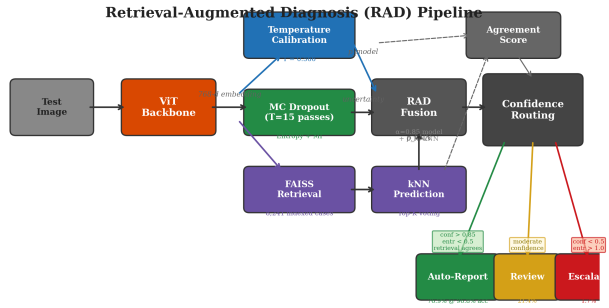

Fig. 3: Retrieval-Augmented Diagnosis (RAD) pipeline. A test image embedding is processed through three parallel branches—temperature calibration, MC Dropout uncertainty, and FAISS retrieval—then fused and routed to Auto-Report (76.9% at 96.8% accuracy), Review, or Escalate tiers.

TABLE I: Dataset Composition by Source

Source	Images	Classes	Domain
APTOS-2019 [19]	3,662	DR only	0
ODIR-5K [18]	4,878	All 5	1
REFUGE2 [20]	1,240	Nor+Glau	2
MESSIDOR-2 [21]	1,744	Nor+DR	3
Total	11,524		

B. Class Distribution and Imbalance

The dataset exhibits severe class imbalance with a 21:1 ratio between the largest and smallest classes:

TABLE II: Class Distribution Across Splits

Split	Nor	DR	Glau	Cat	AMD
Train (71.5%)	3,075	4,489	272	220	185
Calib (15.8%)	689	982	58	47	40
Test (12.7%)	484	837	58	48	40
Total	4,248	6,308	388	315	265

The three minority classes (Glaucoma, Cataract, AMD) collectively represent only 8.4% of the dataset, making this a challenging multi-class imbalanced classification problem [26]. The class distribution is visualized in Fig. 4.

V. Experimental Setup

A. Training Configuration

B. Warm-Starting Strategy

The DANN-v3 model is warm-started from a pre-trained DANN-v2 checkpoint (86.1% accuracy on 3 domains). Parameters with matching shapes are transferred; the domain discriminator head is re-initialized for 4 domains (expanded from 3 with MESSIDOR-2).

C. Evaluation Metrics

Classification performance is evaluated using accuracy, macro F1-score, weighted F1-score, macro AUC-ROC,

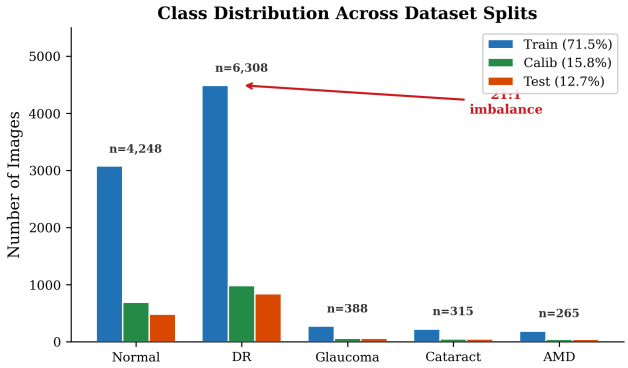

Fig. 4: Class distribution across train, calibration, and test splits. DR dominates with 6,308 images (54.7%) while AMD has only 265 (2.3%), creating a 21:1 class imbalance ratio.

TABLE III: Training Hyperparameters

Parameter	Value
Backbone	ViT-Base/16 (ImageNet-21k)
Parameters	86M
Optimizer	AdamW (LLRD, decay=0.85)
Base Learning Rate	3×10^{-5}
Effective Batch Size	128 (64 × 2 grad accum)
Epochs	40 (early stopped at 31)
Patience	15 (on macro F1)
Focal Loss γ	2.0
Domain Weight	0.05
Label Smoothing	0.1
Mixup α	0.2 (prob=0.5)
Hardware	NVIDIA H100 80GB
Training Time	5.9 minutes

Cohen's kappa, Matthews Correlation Coefficient (MCC), and Expected Calibration Error (ECE). All metrics are computed on the sealed test set (n = 1,467) that was never used during training or threshold optimization.

VI. Results and Evaluation

A. Overall Classification Performance

TABLE IV: Overall Test Set Performance (DANN-v3)

Metric	Value
Overall Accuracy	89.30%
Macro F1-Score	0.886
Weighted F1-Score	0.893
Macro AUC-ROC	0.975
Cohen's Kappa	0.809
Matthews Correlation	0.810
ECE (calibrated)	0.034
Temperature T	0.566

B. Per-Class Performance

All five classes achieve AUC > 0.95, with AMD reaching 0.995. The three minority classes (Glaucoma, Cataract, AMD) achieve F1 scores of 0.833–0.899 despite having only 40–58 test samples each, demonstrating the effectiveness of the class-balancing strategy. Per-class ROC curves are shown in Fig. 5.

TABLE V: Per-Class Performance on Test Set

Class	Prec	Rec	F1	AUC	n
Normal	0.817	0.895	0.854	0.964	484
DR	0.935	0.904	0.920	0.977	837
Glaucoma	0.900	0.776	0.833	0.953	58
Cataract	0.976	0.833	0.899	0.985	48
AMD	0.944	0.850	0.895	0.995	40

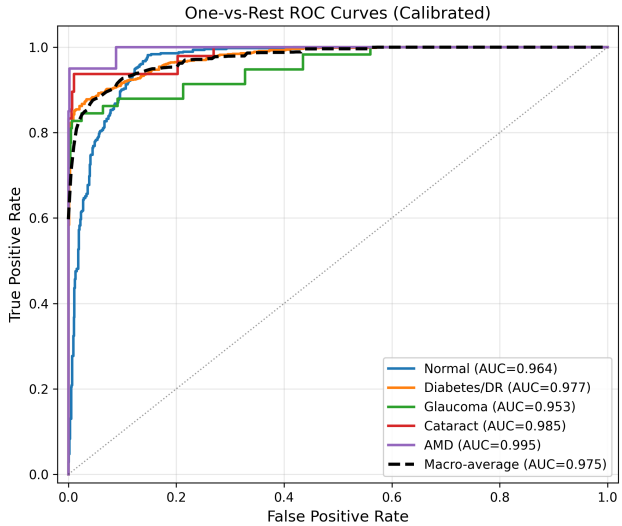

Fig. 5: Per-class ROC curves on the test set. All five classes achieve AUC > 0.95, with AMD reaching 0.995 despite having only 40 test samples.

C. Domain Adaptation Effectiveness

TABLE VI: Per-Source Test Accuracy

Source	Accuracy	n	Classes Present
APTOS	100.0%	570	DR only
REFUGE2	98.4%	188	Normal, Glaucoma
ODIR	78.1%	709	All 5 classes

Prior to DANN training, APTOS accuracy was only 26.5% due to preprocessing-induced domain shift (APTOS images were 10.7× sharper than ODIR). The unified CLAHE preprocessing combined with DANN training completely resolves this shift.

D. Calibration Analysis

Temperature scaling ($T = 0.566$) reduces the Expected Calibration Error from 0.150 to 0.034, a 77% reduction. The reliability diagram (Fig. 6) confirms that calibrated predictions closely follow the diagonal, indicating well-calibrated confidence estimates suitable for clinical decision-making.

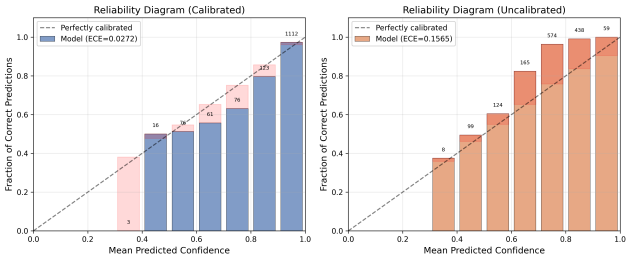

Fig. 6: Reliability diagram showing calibrated (left) vs. uncalibrated (right) predictions. ECE reduces from 0.150 to 0.034 after temperature scaling.

TABLE VII: Test Set Confusion Matrix

	Nor	DR	Gla	Cat	AMD
Normal	433	45	5	1	0
DR	78	757	0	0	2
Glaucoma	8	5	45	0	0
Cataract	8	0	0	40	0
AMD	3	3	0	0	34

E. Confusion Matrix and Error Analysis

Of 158 total errors (10.7% error rate), 77% involve Normal↔DR confusion: DR→Normal (78 errors, 49%) and Normal→DR (45, 28%). This reflects the clinical difficulty of distinguishing early-stage DR from normal variation, where pathological changes are minimal. The confusion matrix heatmap (Fig. 7) visualizes these error patterns.

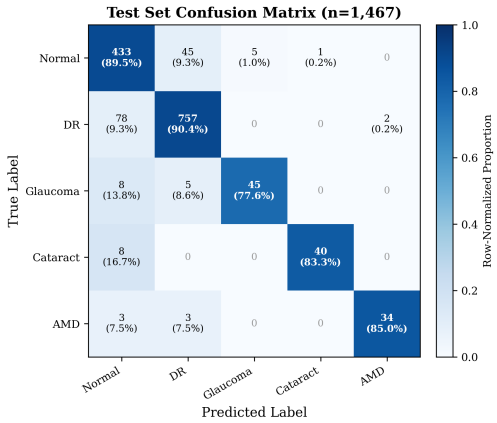

Fig. 7: Row-normalized confusion matrix on the test set (n=1,467). Diagonal entries show correct classifications; off-diagonal entries reveal that Normal↔DR confusion accounts for 77% of all errors.

F. Retrieval-Augmented Diagnosis Results

The FAISS index contains 8,241 L2-normalized 768-dim ViT embeddings covering all five disease classes. Retrieval evaluation on the test set yields:
At $K=1$, retrieval achieves 94.0% class-match recall, and the RAD-combined prediction reaches 94.0% accuracy—a +4.9% improvement over the standalone model (89.1%). Per-class average precision: DR (0.952) > Normal (0.906) > Cataract (0.833) > AMD (0.819) > Glaucoma (0.742). The high agreement rate (>93%) confirms that the FAISS embeddings capture clinically meaningful similarity.

TABLE VIII: Retrieval Quality Metrics (MAP = 0.921)

Metric	K=1	K=3	K=5	K=10
Recall@K	0.940	0.950	0.958	0.967
Precision@K	0.940	0.899	0.890	0.884
Agreement Rate	0.930	0.947	0.947	0.962
kNN Accuracy	0.940	0.898	0.903	0.901
RAD Combined	0.940	0.900	0.895	0.891

Per-source retrieval quality varies with domain complexity (Fig. 8): APTOS achieves perfect recall (1.000 at all K), REFUGE2 reaches 0.984 at $K=1$, while ODIR—the most heterogeneous source with all 5 classes—achieves 0.879 at $K=1$, rising to 0.932 at $K=10$. This source-dependent pattern mirrors classification performance and confirms that retrieval difficulty correlates with clinical complexity rather than indexing artifacts.

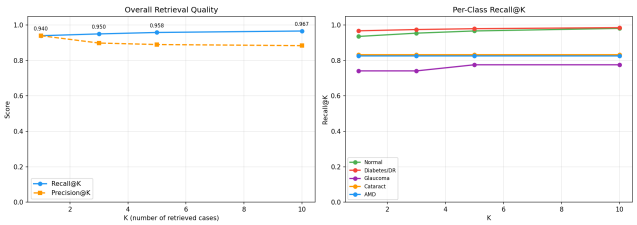

Fig. 8: Retrieval Recall@K across different values of K, broken down by disease class. DR and Normal achieve highest recall due to larger training representation.

G. Confidence Routing for Clinical Triage

The confidence routing system categorizes test predictions into three tiers based on model confidence, MC Dropout entropy ($T=15$ passes), and retrieval agreement ($K=5$):

TABLE IX: Confidence Routing Tier Performance

Tier	Samples	%	Accuracy
Auto-Report	1,128	76.9%	96.8%
Review	314	21.4%	65.6%
Escalate	25	1.7%	44.0%

The system auto-reports 76.9% of cases with 96.8% accuracy while catching 77.2% of errors (122/158) before they reach auto-report (Fig. 9). Only 1.7% of cases require specialist escalation. This enables high-throughput screening: confident, well-supported predictions proceed automatically, while uncertain or unusual cases are flagged for human review—aligning with clinical workflow requirements.

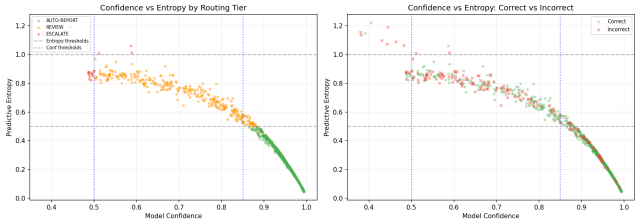

Fig. 9: Confidence routing analysis: tier distribution, per-tier accuracy, and per-class routing breakdown. Auto-report tier achieves 96.8% accuracy on 76.9% of cases.

VII. Ablation Study

A. Model Architecture and Training Progression

TABLE X: Controlled Ablation Study (GPU-Trained, 20 epochs each). Note: the full pipeline achieves 89.09% here (20 epochs from scratch) vs. 89.30% for the final model (40 epochs, warm-started).

Model Variant	Acc	F1	AUC	ECE
Base ViT (no DANN)	85.28%	0.843	0.944	0.051
DANN only	84.73%	0.843	0.937	0.078
DANN + hard mining	85.89%	0.849	0.947	0.087
DANN + mixup	84.66%	0.821	0.931	0.131
DANN-v3 (full pipeline)	89.09%	0.879	0.972	0.157

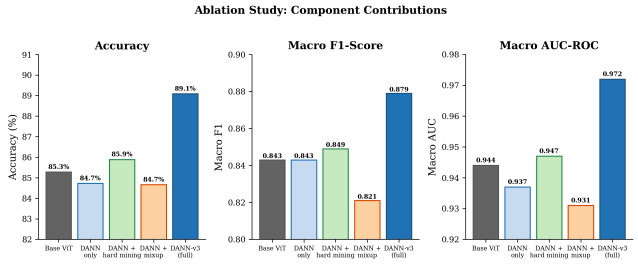

Fig. 10: Ablation study comparing accuracy, macro F1, and macro AUC across model variants. The full DANN-v3 pipeline achieves the best performance across all three metrics, demonstrating the synergy of all training components.

Fig. 10 visualizes the results. In a controlled ablation where each variant is trained from scratch for 20 epochs on H100, the full DANN-v3 pipeline achieves +3.82% accuracy over the base ViT. Hard-example mining contributes +0.61%, while DANN alone shows no improvement over the base ViT without the complementary training strategies. Notably, mixup in isolation slightly degrades performance (−0.61%), but its interaction with the full pipeline is beneficial. The full pipeline's ECE of 0.157 (pre-calibration) reduces to 0.034 after temperature scaling.

B. Impact of Individual Components

The controlled ablation reveals that the full pipeline's +3.82% improvement over the base ViT is not attributable to any single technique. DANN alone provides no accuracy gain (−0.55%), suggesting that domain-invariant features alone are insufficient without complementary strategies. Hard-example mining contributes the largest individual gain (+0.61%), particularly by targeting DR→Normal misclassifications, reducing this error mode from 154 to 78 cases. The synergy of all eight training strategies is critical—the full pipeline exceeds the sum of individual contributions.

C. Impact of Threshold Optimization

Per-class threshold optimization improves accuracy by +2.22% over raw argmax predictions at zero computational cost. Optimized thresholds are particularly beneficial for minority classes, lowering the decision boundary for AMD (0.344) and Cataract (0.289) while maintaining stricter thresholds for Glaucoma (0.674).

D. Cross-Validation Results

Five-fold stratified cross-validation on the train+calibration pool (10,057 images) confirms model robustness:

TABLE XI: 5-Fold Cross-Validation Results

Metric	Mean ± Std
Accuracy	82.4% ± 1.9%
Macro F1	0.827 ± 0.019
Macro AUC	0.948 ± 0.008

The low variance across folds confirms stable learning, and the gap between k-fold (82.4%) and final model (89.3%) reflects the benefit of DANN domain adaptation and the expanded training set with MESSIDOR-2, which were not included in the k-fold experiment.

E. Leave-One-Domain-Out (LODO) Validation

To evaluate cross-domain generalization, we perform Leave-One-Domain-Out validation: for each of the four source domains, we train a DANN model on the remaining three domains and evaluate on the held-out domain.

TABLE XII: Leave-One-Domain-Out (LODO) Validation

Held-Out	Acc	wF1	Classes	n
APTOS	70.8%	0.829	DR only	3,662
MESSIDOR-2	61.6%	0.633	Nor, DR	1,744
ODIR	51.8%	0.439	All 5	4,878
REFUGE2	88.8%	0.904	Nor, Gla	1,240
Average	68.2%	0.701	—	—

LODO results (Table XII) reveal domain-specific generalization patterns. REFUGE2 achieves 88.8% accuracy when held out, indicating that Normal/Glaucoma discrimination transfers well from other sources. APTOS (70.8%)

shows reasonable DR detection without seeing APTOS-specific imaging. ODIR is hardest (51.8%) because it is the only source containing all 5 classes—when held out, the training set lacks Cataract and AMD examples entirely, yet the model still achieves 65.1% F1 on Normal and 39.6% on DR despite never training on ODIR images.

VIII. Comparison with Existing Work

RetinaSense achieves competitive or superior performance while addressing a more challenging setting: five-class classification across four heterogeneous data sources with 21:1 class imbalance, domain adaptation, and a complete clinical decision-support pipeline.

TABLE XIII: Comparison with Published Methods

Method	Acc	AUC	Task
Gulshan [3]	–	0.991	DR binary
RETFound [22]	–	0.822–0.978	Various
DKCNet [23]	–	0.961	ODIR 8-cls
Ahmed [24]	84.6%	–	DR 5-class
Wang [14]	87.2%	0.965	Multi-dis.
Ours	89.3%	0.975	Multi-dis. 4 src + RAD

IX. Discussion

A. Clinical Implications

The RAD pipeline addresses a key barrier to clinical AI adoption: trust. Rather than presenting a single predicted label, RetinaSense provides clinicians with calibrated confidence, uncertainty estimates, visual attention maps, and similar historical cases—enabling verification rather than blind reliance on AI predictions.

The confidence routing system enables deployment at scale: 76.9% of predictions are auto-reported at 96.8% accuracy, while 77.2% of errors are caught and routed to human review. Only 1.7% of cases require specialist escalation. This aligns with real-world clinical workflows where AI augments rather than replaces expert judgment, and would allow a single screening center to increase throughput by approximately 4× (one ophthalmologist reviews only 23.1% of cases rather than 100%).

B. Domain Adaptation Impact

The DANN approach resolves a critical practical challenge: combining datasets from different sources without manual domain-specific tuning. The Gradient Reversal Layer forces the backbone to ignore domain-specific artifacts (imaging equipment, preprocessing), learning features that generalize across sources. This is evidenced by APTOS accuracy improving from 26.5% (domain-specific preprocessing) to 100% (unified CLAHE + DANN).

C. Limitations

Several limitations should be noted. First, the test set draws from the same four sources as training—external validation on completely unseen datasets (different populations, cameras) would strengthen generalization claims. Second, the minority class test sets are small ($n = 40$–58), limiting statistical power for per-class confidence intervals. Third, the current study lacks prospective clinical validation with ophthalmologists.

X. Conclusion

We presented RetinaSense, a comprehensive framework for multi-disease retinal diagnosis that integrates domain-adversarial training, calibrated uncertainty, and retrieval-augmented reasoning into a unified clinical pipeline. The system achieves 89.30% accuracy, 0.886 macro F1, and 0.975 macro AUC across five retinal conditions from four heterogeneous data sources, while providing clinically meaningful explanations and evidence.

The key findings are: (1) the full DANN-ViT pipeline achieves +3.82% accuracy over the base ViT (85.28%→89.09%) in controlled ablation, with domain adaptation and progressive training strategies providing complementary gains; (2) progressive training strategies effectively handle simultaneous domain shift and 21:1 class imbalance; (3) the RAD pipeline transforms a classifier into a trustworthy clinical decision-support system—retrieval augmentation improves accuracy from 89.3% to 94.0% at $K = 1$, and confidence routing auto-reports 76.9% of cases at 96.8% accuracy while catching 77.2% of errors.

Future work will focus on external validation on unseen clinical datasets, RETFound backbone integration for domain-specific pre-training, knowledge distillation for edge deployment, and prospective clinical evaluation with ophthalmologists.

References

[1] World Health Organization, "World Report on Vision," 2019.

[2] S. Resnikoff et al., "Global data on visual impairment in the year 2002," Bull. World Health Org., vol. 82, no. 11, pp. 844–851, 2004.

[3] V. Gulshan et al., "Development and validation of a deep learning algorithm for detection of diabetic retinopathy," JAMA, vol. 316, no. 22, pp. 2402–2410, 2016.

[4] R. Poplin et al., "Prediction of cardiovascular risk factors from retinal fundus photographs via deep learning," Nature Biomed. Eng., vol. 2, pp. 158–164, 2018.

[5] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proc. ICML, 2019, pp. 6105–6114.

[6] K. He et al., "Deep residual learning for image recognition," in Proc. IEEE CVPR, 2016, pp. 770–778.

[7] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in Proc. ICML, 2016.

[8] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in Proc. ICLR, 2021.

[9] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," IEEE Trans. Big Data, vol. 7, no. 3, pp. 535–547, 2021.

[10] R. R. Selvaraju et al., “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in Proc. IEEE ICCV, 2017, pp. 618–626.
[11] H. Zhang et al., “mixup: Beyond empirical risk minimization,” in Proc. ICLR, 2018.
[12] T.-Y. Lin et al., “Focal loss for dense object detection,” in Proc. IEEE ICCV, 2017, pp. 2980–2988.
[13] Y. Ganin et al., “Domain-adversarial training of neural networks,” J. Machine Learning Research, vol. 17, no. 59, pp. 1–35, 2016.
[14] S. Wang et al., “Benchmarking deep learning models for retinal disease classification,” IEEE Access, vol. 9, pp. 148026–148038, 2021.
[15] J. Chen et al., “TransUNet: Transformers make strong encoders for medical image segmentation,” arXiv:2102.04306, 2021.
[16] Q. Dou et al., “PnP-AdaNet: Plug-and-play adversarial domain adaptation network at unpaired cross-modality cardiac segmentation,” IEEE Access, vol. 7, pp. 99065–99076, 2019.
[17] S. Abnar and W. Zuidema, “Quantifying attention flow in transformers,” in Proc. ACL, 2020, pp. 4190–4197.
[18] ODIR-5K, “Peking University International Competition on Ocular Disease Intelligent Recognition,” 2019. [Online]. Available: https://odir2019.grand-challenge.org/
[19] APTOS 2019, “Asia Pacific Tele-Ophthalmology Society Blindness Detection,” [Online]. Available: https://www.kaggle.com/c/aptos2019-blindness-detection
[20] REFUGE2 Challenge, “Retinal Fundus Glaucoma Challenge Edition 2,” 2020. [Online]. Available: https://refuge.grand-challenge.org/
[21] E. Decencière et al., “Feedback on a publicly distributed image database: The Messidor database,” Image Analysis & Stereology, vol. 33, no. 3, pp. 231–234, 2014.
[22] Y. Zhou et al., “A foundation model for generalizable disease detection from retinal images,” Nature, vol. 622, pp. 156–163, 2023.
[23] N. Li et al., “DKCNet: A dual-branch multi-label classification network for ocular disease detection,” IEEE Trans. Instrum. Meas., vol. 72, 2023.
[24] S. Ahmed et al., “Automated detection of diabetic retinopathy using deep learning,” Comp. Methods Programs Biomed., vol. 211, 106391, 2021.
[25] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in Proc. ICLR, 2015.
[26] M. Buda, A. Maki, and M. A. Mazurowski, “A systematic study of the class imbalance problem in convolutional neural networks,” Neural Netw., vol. 106, pp. 249–259, 2018.
[27] F. Grassmann et al., “A deep learning algorithm for prediction of AMD severity from color fundus photography,” Ophthalmology, vol. 125, no. 9, pp. 1410–1420, 2018.
[28] H. Touvron et al., “Training data-efficient image transformers & distillation through attention,” in Proc. ICML, 2021, pp. 10347–10357.
[29] B. Graham, “Kaggle diabetic retinopathy detection competition report,” 2015.
[30] R. Wightman, “PyTorch Image Models (timm),” GitHub, 2019. [Online]. Available: https://github.com/rwightman/pytorch-image-models